LLMs Unplugged | Cybernetic Studio

Australian
National
University

Basic Generation

Use a pre-trained model to generate new text through weighted random sampling.

You will need

- your completed bigram model (i.e. your filled-out grid) from *Basic Training*
- d10 (or similar) for weighted sampling
- pen & paper for writing down the generated "output text"

Your goal

To generate new text from your bigram language model. **Stretch goal**: keep going, generating as much text as possible. Write a whole book!

Key idea

Language models generate text by predicting one word at a time based on learned patterns. Your trained model provides the "next word" options and their relative probabilities; dice rolls provide the randomness to choose one of those options (and this process can be repeated indefinitely).

Algorithm

1. **choose a starting word** — pick any word from the first column of your grid
2. **look at that word's row** to identify all possible next words and their counts
3. **roll dice weighted by the counts** (see the *Weighted Randomness* lesson)
4. **write down the chosen word** and use that as your next starting word
5. **repeat** from step 2 until you reach the desired length *or* a natural stopping point (e.g. a full stop `.`)

Example

Using the same bigram model from the example in *Basic Training*:

<table>
<tr><th></th><th>see</th><th>spot</th><th>run</th><th>.</th><th>jump</th><th>,</th></tr>
<tr><td>see</td><td></td><td>||</td><td></td><td></td><td></td><td></td></tr>
<tr><td>spot</td><td></td><td></td><td>|</td><td></td><td>|</td><td>||</td></tr>
<tr><td>run</td><td></td><td></td><td></td><td>||</td><td></td><td>|</td></tr>
<tr><td>.</td><td>|</td><td></td><td>|</td><td></td><td>|</td><td></td></tr>
<tr><td>jump</td><td></td><td></td><td></td><td>||</td><td></td><td>|</td></tr>
<tr><td>,</td><td></td><td>||</td><td>|</td><td></td><td>|</td><td></td></tr>
</table>

- choose (for example) `see` as your starting word
- `see` (row) ⟶ `spot` (column); it's the only option, so write down `spot` as next word
- `spot` ⟶ `run` (25%), `jump` (25%) or `,` (50%); roll dice to choose
- let's say dice picks `run`; write it down
- `run` ⟶ `.` (67%) or `,` (33%); roll dice to choose
- let's say dice picks `.`; write it down
- `.` ⟶ `see` (33%), `run` (33%) or `jump` (33%); roll dice to choose
- let's say dice picks `see`; write it down
- `see` ⟶ `spot`; it's the only option, so write down `spot`... and so on

After the above steps, the generated text is *"see spot run. see spot"*